Assignment 4

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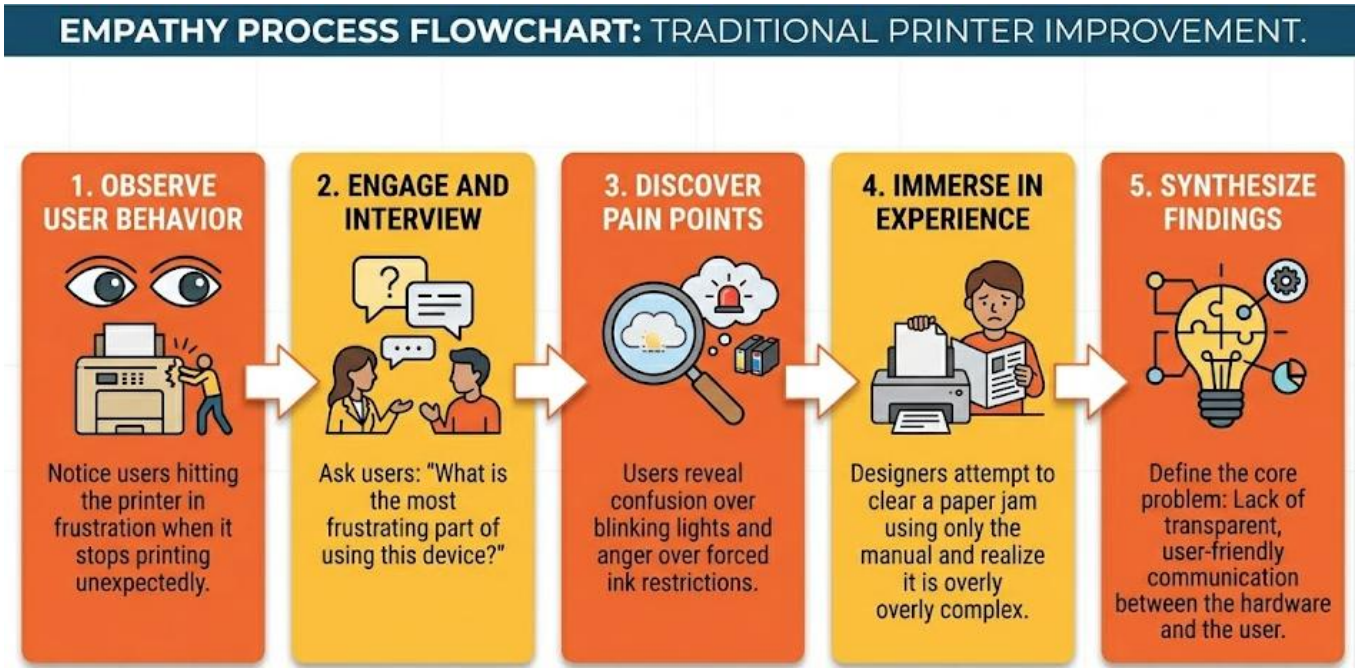
Task 1: Identify and Fix a Product

Product Identified: Traditional Office/Home Printer

Explore its Features: Standard features include document printing, flatbed scanning, wireless network connectivity, and multi-color ink cartridges.

Product Improvement: Standard printers are notorious for ambiguous error codes (e.g., "Error 0x892") and frustrating ink management (stopping a black-and-white print because the yellow ink is low).

Fix: Replace the blinking LED lights with an interactive e-ink display that gives plain-text solutions (e.g., "Paper jam in the rear tray. Click here for a video tutorial"). Additionally, update the firmware so that if a color cartridge is empty, the printer automatically overrides to print in grayscale rather than locking the machine.



Task 2: User Research, Persona & Journey Mapping

Product Concept: "Lumina" - A smart, circadian-rhythm regulating desk lamp designed for remote workers that automatically adjusts color temperature based on the time of day to reduce eye strain.

User Research Conducted: Conducted an online survey of 50 remote IT professionals. *Findings:* 72% reported experiencing severe eye strain or headaches after 6 PM, and 65% reported difficulty falling asleep due to late-night exposure to harsh, blue-light-emitting monitor setups.

Persona Creation:

- **Demographic Information:**
  - **Name:** Sarah Jenkins
  - **Age:** 32 | **Occupation:** Remote Software Engineer | **Location:** Urban Apartment
- **Goal and Objectives:**
  - Reduce daily eye strain and end-of-day migraines.
  - Improve sleep hygiene without having to log off work earlier.
  - Maintain a productive, focused workspace environment.
- **Psychographic Information:**
  - **Interests:** Minimalist workspace setups, health/wellness tech, productivity podcasts.
  - **Personality Traits:** Detail-oriented, health-conscious, willing to invest in high-quality tools.
- **Behaviour and Preference:**
  - Works long, irregular hours (often late into the night).
  - Dislikes bright overhead lighting; prefers focused desk lamps.
  - Relies on automated smart home tech (Alexa/Google Home).
- **Challenges and Pain Points:**
  - Chronic dry eyes and headaches from screen glare.
  - Disruptive sleep patterns leading to grogginess the next day.
  - Forgets to manually adjust monitor brightness as the sun goes down.

User Journey (Brief): Sarah works late ➔ Eyes start hurting around 7 PM ➔ Tries using "dark mode" on apps, but the room lighting is still harsh ➔ Searches online for "eye-strain relief desk setups" ➔ Discovers Lumina ➔ Purchases and integrates it ➔ Lamp automatically dims and shifts to warm amber at night ➔ Sarah finishes work comfortably and falls asleep faster.

Task 3: Waterfall Development Model

Scenario: Developing the "Lumina" Smart Lamp Companion Mobile App for the startup company.

Waterfall Development Model for Lumina App:

1. **Requirements Gathering & Analysis:** * *Action:* Document all necessary app features. The app must connect via Bluetooth, allow users to set their sleep/wake schedule, and feature a manual override color-wheel. Determine server requirements for user data storage.
2. **System Design:**
    - *Action:* UI/UX team creates screen mockups (wireframes). Backend developers map out the database architecture for saving custom lighting presets.
3. **Implementation (Coding):**
    - *Action:* The software engineers write the actual code. They build the iOS and Android versions separately, integrating the Bluetooth API to communicate with the lamp's hardware.
4. **Testing:**
    - *Action:* Quality Assurance (QA) team tests the app. They check for bugs, ensure the lamp changes color exactly when the app schedule dictates, and verify that the Bluetooth connection doesn't drop.
5. **Deployment:**
    - *Action:* The app is officially launched on the Apple App Store and Google Play Store simultaneously with the physical release of the lamp.
6. **Maintenance:**
    - *Action:* Pushing out patches for bugs reported by users (e.g., fixing a crash on newer OS updates) and maintaining server stability.

Task 4: Customer Journey Map (Tool Framework)

CUSTOMER JOURNEY MAP: LUMINA SMART DESK LAMP

PERSONA: Sarah Jenkins (32, Remote Software Engineer, experiences eye strain)

	AWARENESS	CONSIDERATION	PURCHASE	ONBOARDING	DAILY USE & LOYALTY
USER ACTIONS	• Experiencing eye fatigue; Googles "screen headaches."	• Comparisng eye tech vs vertw comparison.	• Shops now purchase. Lumina Smart Desklop.	• Start lumia Smart lamp and forium boeartian.	• Daily use & loyalaion; provide health community.
TOUCHPOINTS	• Engtancing eye fatigue; googes ond websites.	• YouTube reviews, tech blogs, Lumina website.	• Purchase price & shopping detiails.	• Mobile app proaclive to launor your omboarding.	• Daily use daily insights and confortability.
USER EMOTIONS	• Excited and confident in choice.	• Images and nonours experts.	• Excited and confident in choice.	• Excited and confident in choice.	• Excited and confident in choice.
PAIN POINTS	• Experiencing eye fatigue; Googles "screen headaches."	• Reduces pairing struggles; screen headators.	• Bluetooth pairing forions app interface.	• Bluetooth pairing struggles; confusing app interface.	• Introduce referral program; provide health insights based on usage.
OPPORTUNITIES	• Introduce referral program; provide health health inots on usage.	• Introduce referral program; provide health insights usage.	• Introduce referral program; provide health insights.	• Introduce referral program; provide health insights.	• Introduce referral program; provide health insights based on usage.